

2(a)	<i>any one from:</i> 1 industrialisation ; 2 transportation ; 3 power stations ; 4 burning of fuels ;	1
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Question	Answer	Marks																																			
2(b)	any three from: 1 genetically modified has a higher (dry) mass throughout the experiment ; ora 2 genetically modified has a (slightly) higher rate of / steeper, increase (in dry mass) ; ora 3 the largest difference in (dry) mass is after day 8 ; 4 paired data quote with units ; <table><tr><th rowspan="2">time / days</th><th colspan="3">dry mass / g dm⁻³</th></tr><tr><th>genetically modified ± 0.02</th><th>unmodified ± 0.02</th><th>difference between the two ± 0.04</th></tr><tr><td>2</td><td>0.22</td><td>0.13</td><td>0.09</td></tr><tr><td>4</td><td>0.48</td><td>0.34</td><td>0.14</td></tr><tr><td>6</td><td>0.62</td><td>0.48</td><td>0.14</td></tr><tr><td>8</td><td>0.88</td><td>0.74</td><td>0.14</td></tr><tr><td>10</td><td>1.24</td><td>0.96</td><td>0.28</td></tr><tr><td>12</td><td>1.54</td><td>1.31</td><td>0.23</td></tr><tr><td>14</td><td>1.86</td><td>1.52</td><td>0.34</td></tr></table> rate of growth over 14 days genetically modified = $1.86 / 14 = 0.13 \text{ g dm}^{-3} \text{ day}^{-1}$ unmodified = $1.52 / 14 = 0.11 \text{ g dm}^{-3} \text{ day}^{-1}$	time / days	dry mass / g dm ⁻³			genetically modified ± 0.02	unmodified ± 0.02	difference between the two ± 0.04	2	0.22	0.13	0.09	4	0.48	0.34	0.14	6	0.62	0.48	0.14	8	0.88	0.74	0.14	10	1.24	0.96	0.28	12	1.54	1.31	0.23	14	1.86	1.52	0.34	3
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2(c)	any three from: 1 (more rubisco so) greater rate of / more, carbon (dioxide) fixation ; AW 2 (so) greater rate of / more, GP produced ; 3 so greater rate of / more TP, produced (from GP) ; 4 (so) greater rate of / more, regeneration of RuBP ; 5 (so) greater rate of / more, Calvin cycle ; 6 AVP ; e.g. carbon fixation is a rate-limiting step / concentration of rubisco is a limiting factor	3
2(d)	any three from: 1 glucose for respiration ; 2 starch for storage ; 3 cellulose to make cells walls ; 4 sucrose for, translocation / described ; 5 fatty acids and glycerol / lipid, to make membranes or fatty acids and glycerol / lipid / fats, for storage or fatty acids to make acetyl CoA (for Krebs cycle) ; 6 amino acids to make, proteins / enzymes ; 7 proteins for, growth / repair ;	3

Question	Answer	Marks
2(e)	<i>any one from:</i> 1 <i>idea of fast(er) growth rate / reproduction ;</i> 2 <i>quicker to set up / AW ;</i> 3 <i>cheaper to set up / AW ;</i> 4 <i>(as aquatic) not using (useful) land ;</i> 5 <i>can, culture / grow, algae in the lab ;</i>	1

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